

A two-state component for cyclic four-flows on even Möbius ladders

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Abstract

For an even integer $r \geq 4$, let M_{2r} be the Möbius ladder consisting of a cycle on $2r$ vertices and the matching joining opposite vertices. Consider its nowhere-zero $\mathbb{Z}/4\mathbb{Z}$ -flows, with two flows adjacent when the support of their difference is a simple cycle. We prove that this reconfiguration graph is disconnected for every such r . An alternating rim flow and its negative form a component of exactly two states, and an explicit third flow lies outside that component. The proof applies the known obstruction from the three nonzero signed cycle values: equal-parity chord endpoints force it on every cycle containing a chord. Thus the set of even parameters with connected flow reconfiguration graph is empty.

1 Statement and context

Fix an orientation D of a graph G . A nowhere-zero $\mathbb{Z}/4\mathbb{Z}$ -flow is a function $f : E(G) \rightarrow \{1, 2, 3\}$ for which, at every vertex, the sum on outgoing edges minus the sum on incoming edges is zero modulo four. The vertices of $\mathcal{F}(G, \mathbb{Z}/4\mathbb{Z})$ are these flows. Distinct flows f, h are adjacent exactly when

$$\text{supp}(h - f) = \{e \in E(G) : h(e) \neq f(e)\}$$

is the edge set of a simple cycle. All simple cycles are allowed. Reversing an edge of D and negating the value on that edge gives an isomorphism between the corresponding reconfiguration graphs: it preserves nonzero values, conservation, and difference supports. Connectivity is therefore independent of D .

For even $r \geq 4$, the graph M_{2r} has vertex set $\mathbb{Z}/(2r)\mathbb{Z}$, rim edges $\{i, i+1\}$, and chords $\{j, j+r\}$ for $0 \leq j < r$. Each undirected edge occurs once, so the graph has $2r$ vertices and $3r$ edges. Use the orientation

$$e_i = (i, i+1) \quad (0 \leq i < 2r), \quad c_j = (j, j+r) \quad (0 \leq j < r), \quad (1)$$

where rim indices are taken modulo $2r$.

Theorem 1. *For every even integer $r \geq 4$, the graph $\mathcal{F}(M_{2r}, \mathbb{Z}/4\mathbb{Z})$ is disconnected. More precisely, in the orientation (1), the flows*

$$f_+(e_i) = \begin{cases} 1, & i \text{ even,} \\ 3, & i \text{ odd,} \end{cases} \quad f_+(c_j) = 2, \quad f_- = -f_+ \quad (2)$$

form a connected component of exactly two states. The flow defined in (3) lies outside this component. Consequently,

$$S := \{r \geq 4 : r \text{ is even and } \mathcal{F}(M_{2r}, \mathbb{Z}/4\mathbb{Z}) \text{ is connected}\} = \emptyset.$$

Esperet, Hendrey, Lagoutte, Marseloo, Norin, and Steiner [2] study nowhere-zero flow reconfiguration; their Section 2.4 exhibits the obstruction from three nonzero signed values on a cycle and paired components for K_4 . Cranston, Li, Su, Wang, and Xu [1] prove $\mathbb{Z}/4\mathbb{Z}$ -flow connectivity for cubic bipartite graphs (Theorem 3.2 and the following paragraph), disconnection for three-edge-colorable cubic graphs containing a triangle (Theorem 3.3), and connectivity for prisms (Theorem 3.4).

The even Möbius ladders considered here are cubic and three-edge-colorable: alternate two colors around the even rim and use a third on the chord matching. They are nonbipartite, since a chord and an r -edge rim path form an odd cycle. They are also triangle-free. Indeed, a triangle could use at most one chord, and its other two edges would give a rim path of length two between opposite vertices, impossible for $r \geq 4$. Thus the bipartite and triangle criteria above do not decide this family. A prism on $2r$ vertices is bipartite for even r , so the prism theorem does not identify the present graphs either.

The contribution here is the complete even-parameter classification and the explicit component, using the known signed-value obstruction. The uniform step is that the endpoints of every chord have the same parity; this blocks every potential update using any chord. The comparison with prior work is restricted to the cited versions and a bounded literature search; no unrestricted priority claim is made. As adjacent context, edge-graceful labellings of trees also impose vertex-sum conditions in cyclic groups [3]. That reference concerns a different labelling problem and is not used in the proof below.

2 The closed pair

All flow arithmetic in the proof is in $\mathbb{Z}/4\mathbb{Z}$. For a traversal of an edge, its *signed value* is its flow value if the traversal agrees with D , and the negative of that value otherwise.

Lemma 1. *If distinct flows differ only on a simple cycle C , then, in a cyclic traversal of C , their difference has one constant nonzero signed value $\lambda \in \{1, 2, 3\}$ on every edge. Conversely, adding such a constant along a cyclic traversal preserves conservation, and is a legal move exactly when every resulting edge value is nonzero. In particular, no move on C is legal from a flow whose signed values on C include 1, 2, and 3.*

Proof. At each vertex of C , conservation of the difference equates its signed incoming and outgoing values. Going around C shows that the value is constant. Since the flows are distinct, this constant is nonzero, so the difference support is the whole cycle. This also explains the equivalence with the convention that the difference is supported within a cycle. A constant cycle increment cancels at every vertex. It is legal precisely when no signed value becomes zero. If the values 1, 2, 3 all occur, then each possible nonzero increment λ cancels an edge with signed value $-\lambda$. $\square$

First, (2) defines two distinct nowhere-zero flows. At any vertex i the rim contribution to outgoing minus incoming flow is

$$f_+(e_i) - f_+(e_{i-1}) = 2.$$

The chord contribution is 2 or -2 , both equal to 2 modulo four. The total vanishes, including at vertices 0 and r . Negation gives the same conclusion for f_- . Their values on e_0 are distinct.

For f_+ , every rim traversal from an even vertex to an odd vertex has signed value 1, and every rim traversal from odd to even has signed value 3. This follows directly from the alternating definition, including the rim edge across the index wrap. For f_- the roles of 1 and 3 are interchanged.

Consider any simple cycle C containing a chord, and traverse a chosen chord from u to v . The edges immediately before and after this chord are both rim edges, because the chords form

a matching. Since r is even, u and v have the same parity. The preceding rim traversal and the following rim traversal therefore have opposite parity directions. Their signed f_+ -values are 1 and 3 in some order, while the chord has signed value 2 in either direction. Lemma 1 blocks the update. The same argument, exchanging 1 and 3, blocks it from f_- . This covers cycles of every length and with any number of chords.

A simple cycle containing no chord must be the entire rim. In a cyclic traversal its signed values include both 1 and 3, so increments 1 and 3 are forbidden. Increment 2 is legal and exchanges f_+ with f_- , leaving all chord values unchanged. Each state consequently has exactly the other as its neighbor. The pair is a connected component; the two states have not been identified by negation or any other symmetry.

3 A flow outside the pair

Write $g(e_i) = a_i$ and $g(c_j) = b_j$, and define

$$a_i = \begin{cases} 2, & i \text{ even,} \\ 1, & i \text{ odd and } 0 \leq i < r, \\ 3, & i \text{ odd and } r \leq i < 2r, \end{cases} \quad b_j = \begin{cases} 1, & j = 0 \text{ or } j \text{ odd,} \\ 3, & j \text{ even and } 2 \leq j < r. \end{cases} \quad (3)$$

Every edge value is nonzero. Conservation at a first-half vertex $0 \leq i < r$ is $a_i - a_{i-1} + b_i = 0$, while at a second-half vertex $r \leq i < 2r$ it is $a_i - a_{i-1} - b_{i-r} = 0$. Table 1 checks all cases, including both boundary vertices, using ordinary integer representatives.

Table 1: Conservation for the outside flow; each sum is zero modulo four.

Vertex	Outgoing minus incoming	Integer sum
$i = 0$	$2 - 3 + 1$	0
$1 \leq i < r, i \text{ odd}$	$1 - 2 + 1$	0
$2 \leq i < r, i \text{ even}$	$2 - 1 + 3$	4
$i = r$	$2 - 1 - 1$	0
$r < i < 2r, i \text{ odd}$	$3 - 2 - 1$	0
$r < i < 2r, i \text{ even}$	$2 - 3 - 3$	-4

Because r is even, these cases exhaust the vertices. Hence g is a nowhere-zero flow. Its value $g(e_0) = 2$ differs from both $f_+(e_0) = 1$ and $f_-(e_0) = 3$, so the two-state component is proper. This proves disconnection for every even $r \geq 4$, including $r = 4$. The orientation isomorphism described above proves the result for any fixed orientation, completing the proof of Theorem 1.

4 Exact checks and scope

The argument above is a proof for all the stated parameters and requires no computation. Supplemental checks used exact integer arithmetic modulo four. Complete state enumeration at $r = 4, 6, 8$, followed by testing each unordered pair for connected degree-two difference support, gave Table 2. A separate implementation using spanning-tree leaf elimination recovered the same states and adjacencies.

An additional implementation enumerated every simple cycle at $r = 3, 4, 5, 6, 7, 8, 10, 12, 18$ using binary sums of fundamental cycles and retaining connected degree-two supports. It tested all three

Table 2: Complete finite reconfiguration graphs for three parameters.

r	Graph vertices	Flow states	Reconfiguration edges	Component sizes
4	8	18	21	2, 8, 8
6	12	66	163	2, 64
8	16	258	969	2, 256

increments from each of f_+ and f_- : 268 471 cycles and 1 610 826 possible increments in total. At each tested even parameter, the rim increment by 2 was the only legal move from either state. At the odd controls $r = 3, 5, 7$, additional legal moves left the pair, as expected when the chord endpoints have opposite parity. Conservation and distinctness of all three witness flows were also checked for every even r from 4 through 200, and for $r = 1000, 10000$; these larger tests did not enumerate cycles.

These finite observations do not classify the components outside the displayed pair for general r . No such classification is asserted, nor is an odd-parameter connectivity theorem proved here. The explicit outside state makes any convention about empty state graphs irrelevant. The accompanying reproducibility material records the exact checks and their coverage.

Computational and AI assistance. OpenAI Codex assisted with proof development, literature comparison, Python verification programs, and manuscript preparation. A separate model-context pass reviewed the mathematical argument and produced the spanning-tree and cycle-space checks reported above. This was an additional computational and model review, not external peer review or a claim of statistical independence. The mathematical justification is the self-contained proof in this article; finite computation is supplemental, and no proof-assistant formalization is claimed.

References

- [1] D. W. Cranston, J. Li, B. Su, Z. Wang, and N. Xu, *Reconfiguration of Nowhere-zero Flows*, arXiv:2606.24685v1 (2026). <https://arxiv.org/abs/2606.24685v1>.
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- [3] Z. Zeng, *Odd-Order Trees with Exactly Three Vertices of Degree Two Are Edge-Graceful*, SSRN preprint (2026). <https://doi.org/10.2139/ssrn.7387319>.